DIPLOMA PROGRAMME PRACTICE · SL · SET A

Biology / Paper 2

Question booklet · 50 marks · 90 minutes

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams. Answer all of Section A and one of two Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY

QUESTION 1 [12]

Shell variation across a town

The photographs show examples of colour and banding. In a new survey, researchers counted yellow and non-yellow adult snails at 12 urban and 12 woodland sites. Each site contributed 20 snails. Temperatures represent mean afternoon ground temperature. The counts are pooled across sites.

Habitat	Yellow	Non-yellow	Mean ground temperature / °C
Urban	150	90	28.4
Woodland	96	144	23.1



Three shell phenotypes. Supplied Biology HL Paper 2 review, p. 5. The photographs illustrate categories; the numerical survey is constructed.

(a) Calculate the percentage of yellow snails in each habitat and express the difference in percentage points. **[3]**

(b) Distinguish a colour category from a banding category using the photographs. Why would recording colour alone lose information? **[2]**

Paper 2 · Question 1 (continued)

(c) Propose a selection mechanism that could produce the urban pattern. State a necessary condition for a response across generations. **[3]**

(d) Why would treating all 240 snails in one habitat as independent temperature replicates exaggerate the evidence? Propose a site-level analysis. **[2]**

(e) Design a comparison that tests a thermal effect of colour while controlling a plausible confounder. **[2]**

SECTION A · COMPULSORY
QUESTION 2 **[8]**

Carbon uptake and cell respiration

An algal suspension in light produces a net 2.4 mg O_2 h^{-1}. In darkness the same amount of algae consumes 0.6 mg O_2 h^{-1}. Assume comparable conditions and a constant respiration rate.

(a) Calculate gross oxygen production in light. Explain the need for the dark measurement. **[3]**

(b) Explain how the chloroplast and mitochondrion contribute to carbon cycling within these cells. **[3]**

(c) Give a reason the dark correction could be biased and a control for an abiotic change in measured oxygen. **[2]**

SECTION A · COMPULSORY
QUESTION 3 [7]

Transport from lung to artery

At rest, a student's stroke volume is 68 cm³ beat⁻¹ and heart rate is 72 beats min⁻¹. During steady exercise these increase to 95 cm³ beat⁻¹ and 132 beats min⁻¹. Use cardiac output = stroke volume × heart rate. In a simplified tracer study, the first detections after a pulse at the lung capillaries were: left ventricle, 2.2 s; aorta, 2.7 s; pulmonary vein, 0.8 s; left atrium, 1.4 s. The mitral valve connects the two left chambers and the aortic valve connects the ventricle to the aorta.

(a) Calculate cardiac output in dm³ min⁻¹ for both conditions. **[2]**

(b) Use the detection times to reconstruct the tracer's route from the lung capillaries to the aorta. Identify where each of the two valves is crossed. **[3]**

(c) Explain why elevated cardiac output alone cannot establish that muscle oxygen consumption rose by the same factor. **[2]**

SECTION A · COMPULSORY
QUESTION 4 [7]

From coding sequence to phenotype

A coding strand segment changes from 5′-GCT TTT GAA-3′ to 5′-GCT TTC GAA-3′. Use the mRNA codon assignments GCU=Ala, UUU=Phe, UUC=Phe and GAA=Glu.

(a) Give the original mRNA sequence and amino acid sequence with direction of reading indicated. **[2]**

(b) Identify the mutation type at the nucleotide level and its effect on the polypeptide segment. **[2]**

(c) Compare the likely consequence of deleting the first T of the original TTT triplet. Explain why DNA sequence alone cannot establish the size of a phenotypic effect. **[3]**

SECTION B · EXTENDED RESPONSE
QUESTION 5 [16]

Keeping pathogens out and remembering infection

A respiratory pathogen enters through mucosal surfaces. Some previously exposed individuals develop a faster specific response on re-exposure.

(a) Outline three ways barriers reduce entry or establishment of pathogens. **[3]**

(b) Explain how a phagocyte helps contain microbes that have crossed a barrier. **[5]**

(c) Explain a specific antibody response to an antigen and why re-exposure can produce a faster response. **[7]**

[1] Overall quality of the extended response.

SECTION B · EXTENDED RESPONSE
QUESTION 6 [16]

A woodland after a storm

A storm opens a gap in a woodland canopy and leaves fallen branches. Consider both living organisms and dead organic matter.

(a) Outline how three different feeding roles obtain organic matter in the woodland. **[3]**

(b) Explain why energy available to a secondary consumer is less than the chemical energy originally captured by producers. **[5]**

(c) Discuss how carbon moves between the atmosphere, fallen wood, decomposers and regrowing trees over time. **[7]**

[1] Overall quality of the extended response.